ISTA 322 Final Project

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Motivation & Goal

- Looking at how population of a city is connected with ratings of restaurants
  - Italian restaurants in California
- Gathered information from Yelp and the California census
- Yelp API & CSV with California population information by city
- Finding a correlation between the two data sets
- Using the data to determine other information about California demographics and restaurants

Data Collection & Challenges

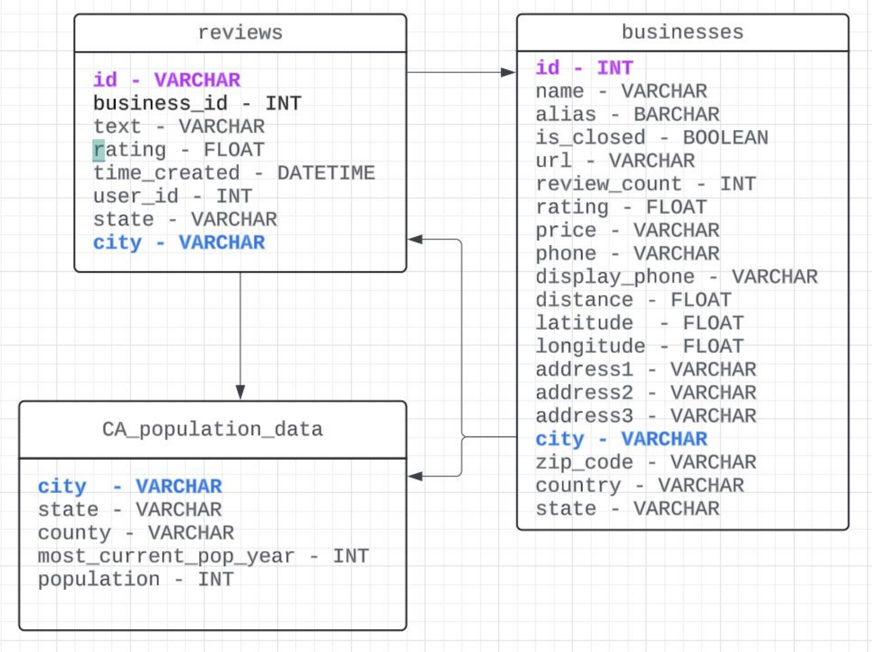
- Gathered data from Yelp API & CSV
- Importing data from Drive
- Merging data
- SQL

Data Cleaning & Transformation

- Challenges with cleaning:
  - Dealing with missing data from the datasets
  - Removing duplicate or unnecessary entries
- Data Transformation
  - Had to extract relevant information
  - Converting data types
  - Normalizing data
- Pandas for data manipulation
- Improves data quality for an accurate analysis
- Had to deal with large data sets and ensure data efficiency and accuracy

Data Schema

- Tables: businesses, reviews, CA_population_data
    - Businesses to reviews: each business can have multiple reviews
    - Businesses to CA_population_data: both tables relate to location information
    - Reviews to CA_population_data: both tables contain city information
- Attributes



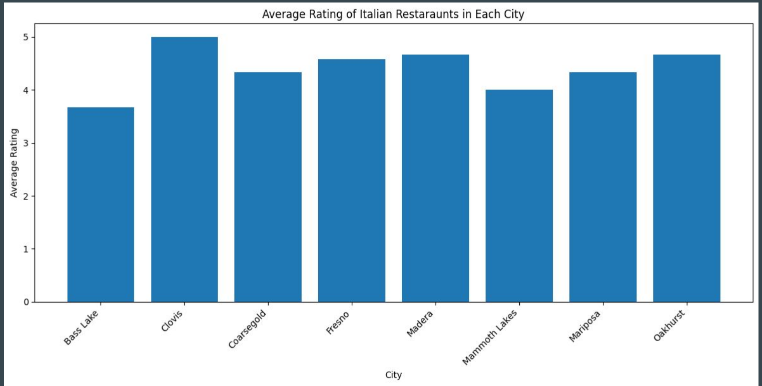
Query 1

Finding the most populated city in California

```python
query1 = """SELECT city, population
FROM CA_population_data
WHERE population < 39356104
ORDER BY population DESC
LIMIT 1;"""
run_query(query1)
```

	city	population
0	Los Angeles city	3881041

Query 2



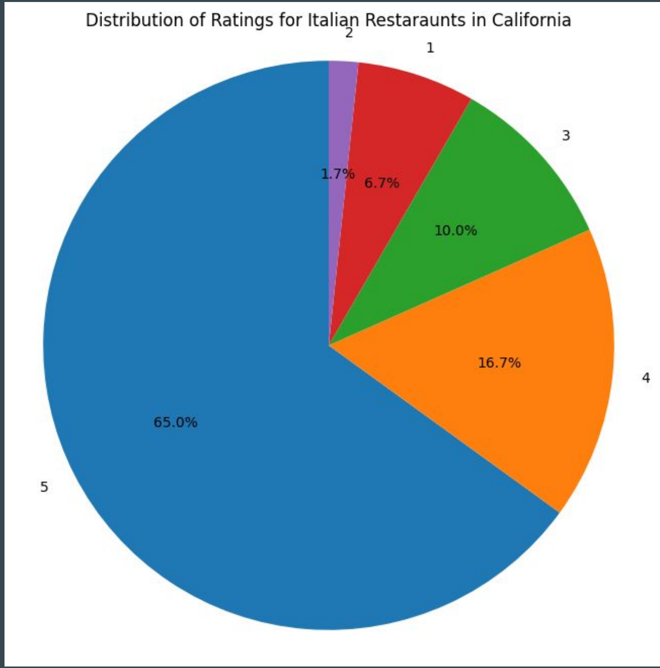
Average rating of italian restaurants in each city

```
# Calculate the average rating for each city
query2 = """SELECT city, AVG(rating) AS average_rating
FROM reviews
GROUP BY city;"""
run_query(query2)
average_ratings = italian_shops_reviews_df.groupby('city')['rating'].mean().reset_index()

# Create a bar graph
plt.figure(figsize=(12, 6))
plt.bar(average_ratings['city'], average_ratings['rating'])
plt.xlabel('City')
plt.ylabel('Average Rating')
plt.title('Average Rating of Italian Restaurants in Each City')
plt.xticks(rotation=45, ha='right')
plt.tight_layout()
plt.show()
```

Query 3

Distribution of Ratings for Italian Restaurants in California

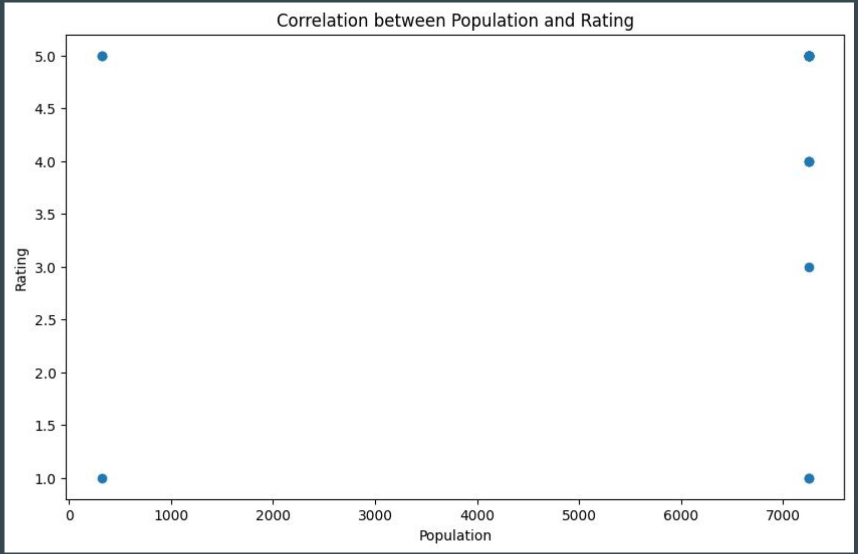
```
# Count the number of Italian restaurants for each rating
query3 = """
SELECT rating, COUNT(id) AS shop_count
FROM businesses
GROUP BY rating;
"""
run_query(query3)
rating_counts = italian_shops_reviews_df['rating'].value_counts()
# Create a pie chart
plt.figure(figsize=(8, 8))
plt.pie(rating_counts, labels=rating_counts.index, startangle=90, autopct='%1.1f%%')
plt.axis('equal')  # Equal aspect ratio ensures that pie is drawn as a circle
plt.title('Distribution of Ratings for Italian Restaurants in California')
plt.show()
```



Query 4

Correlation between population and rating

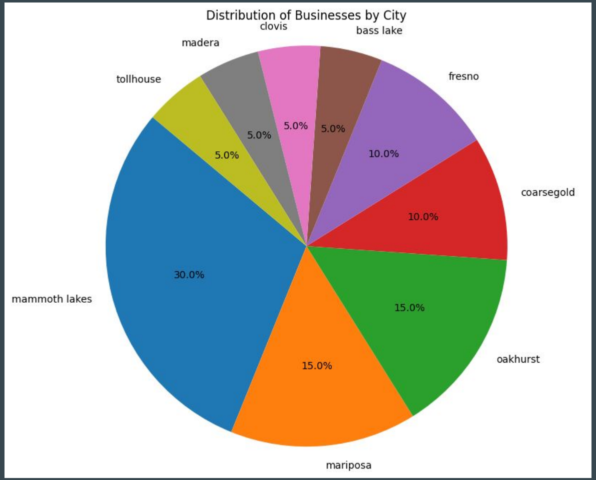
```
query_4 = """SELECT p.population, AVG(r.rating) AS average_rating
FROM ca_population p
JOIN reviews r ON p.city = r.city
GROUP BY p.population;"""
run_query(query_4)

plt.figure(figsize=(10, 6))
plt.scatter(extracted_data['population'], extracted_data['rating'])
plt.xlabel('Population')
plt.ylabel('Rating')
plt.title('Correlation between Population and Rating')
plt.show()
```



Query 5

Distribution of businesses by city

```
# SQL query
query5 = """
SELECT city, COUNT(*) AS business_count
FROM reviews
GROUP BY city;
"""
run_query(query5)
city_counts = italian_shops_reviews_df['city'].value_counts()
# Plotting
plt.figure(figsize=(10, 8))
plt.pie(city_counts, labels=city_counts.index, autopct='%1.1f%%', startangle=140)
plt.axis('equal')  # Equal aspect ratio ensures that pie is drawn as a circle.
plt.title('Distribution of Businesses by City')
plt.show()
```



Outcome/Summary

- Los Angeles is the most populated city in California
- Most Italian restaurants fall within the 3.4-4.5 rating range
- No significant correlation between a city's population and the average rating of its Italian restaurants